Zone-wise Hourly Demand Forecasting on NYC Taxi Trips

Temporal Feature Engineering and Ensemble Regression Approach

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Abstract

This project presents a scalable framework for hourly taxi demand forecasting using 31.86 million NYC Yellow Taxi trip records. Trips were aggregated at the zone-hour level and modeled independently across 25 high-volume zones. Temporal feature engineering including short-term lags, daily seasonality, and rolling averages was applied. Random Forest regression models achieved an average out-of-sample R^2 of 0.92, with best and worst zone performances of 0.9676 and 0.8031 respectively. Results demonstrate that structured temporal features alone can produce strong generalization performance across heterogeneous demand profiles.

1 Problem Statement

Accurate short-term taxi demand forecasting is critical for fleet allocation, surge pricing, and driver positioning. The objective of this project is to forecast hourly pickup demand for individual NYC taxi zones using only historical pickup timestamps and location identifiers.

2 Dataset Description

- Dataset: NYC Yellow Taxi Trip Records
- Total raw trips: 44.42 million
- Time span: 11 months (Jan to Nov 2025)

3 Data Cleaning and Validation

Developed an object-oriented data preprocessing pipeline to clean and standardize individual NYC Taxi datasets. Additionally, built a modular aggregation framework to merge multiple datasets while systematically identifying and handling data anomalies. The YellowTaxiCleaner object has the following methods:

- load_data()
- basic_cleaning()
- feature_engineering()

- summary()
- anomaly_report()
- save_data()
- run()

Anomaly	Count	Percentage
missing_values	10416412	23.45%
zero_distance	1250302	2.81%
zero_fare	2819518	6.35%
zero_passenger	241235	0.54%
invalid_time_order	488283	1.10%
too_short_trip	990168	2.23%
too_long_trip	17145	0.04%
too_slow	1633815	3.68%
too_fast	483841	1.09%
total_flagged_rows	12555678	28.27%
total_rows	44417596	

Figure 1: Anomaly report generated during cleaning

4 Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand temporal demand patterns and statistical characteristics of zone-level trip counts. The cycle plot reveals clear periodic structure, indicating consistent intra-day and weekly demand seasonality. This justifies the inclusion of temporal features in the predictive modeling stage.

Additionally, distributional analysis of trip counts indicates right-skewed behavior, particularly during peak demand periods. This suggests the presence of extreme events and motivates careful residual diagnostics and bias assessment during model evaluation.

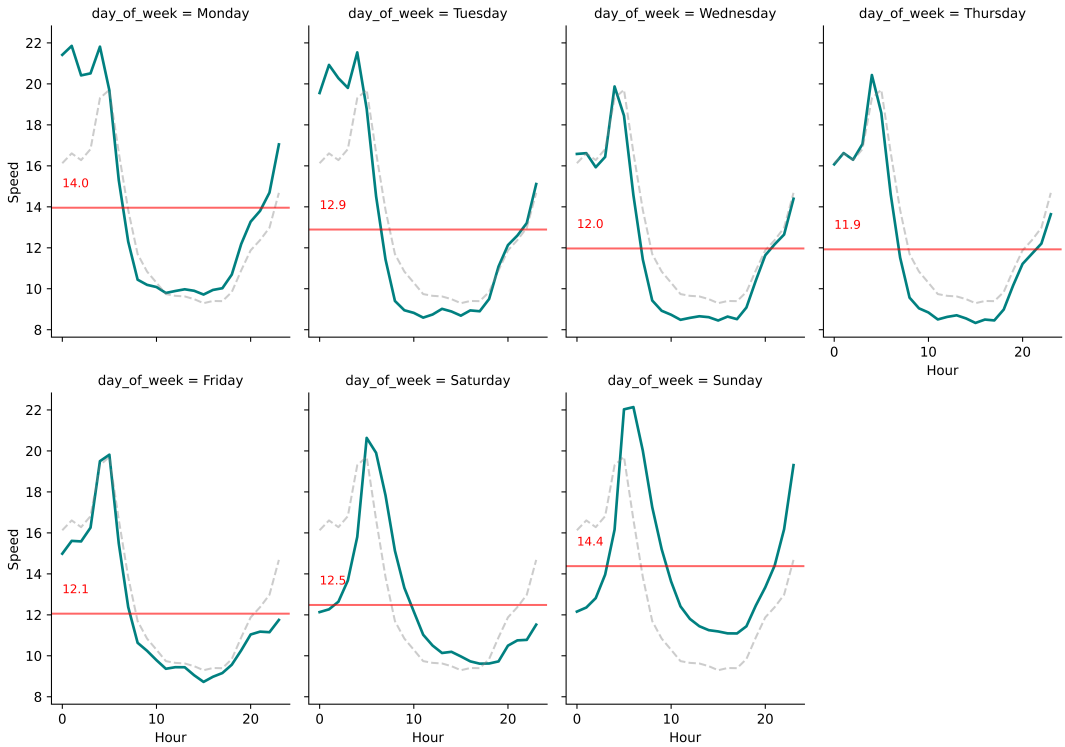

Figure 2: Cycle plot

5 Data Aggregation

Initial features used:

- Pickup datetime
- Pickup Location ID

Data was aggregated to the zone-hour level, reducing dimensionality while preserving temporal structure.

Trips were grouped by:

- Pickup Location ID
- Hourly timestamp (floored to hour)

This reduced the dataset to approximately 800,000 zone-hour observations.

6 Feature Engineering

Temporal features were constructed as follows:

6.1 Short-Term Momentum

- Hourly lag: previous hour demand

6.2 Daily Seasonality

- 24-hourly lag
- Hour of day
- Day of week
- Weekend flag

6.3 Trend Smoothing

- 3-hourly rolling mean demand
- 6-hourly rolling mean demand

All lag features were constructed post-aggregation to preserve temporal causality.

7 Modeling Approach

Independent Random Forest Regressors were trained for each of 25 high-volume zones.

- Train-test split: chronological (Jan to Sept for training and Oct-Nov for testing)
- Target variable: hourly trip count
- Evaluation performed on out-of-sample test set

Zone-wise modeling was chosen to:

- Capture heterogeneous demand behavior
- Improve parallelization potential
- Avoid artificial ordinal encoding of location IDs

8 Evaluation Metrics

Performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 score
- Relative MAE (MAE divided by mean demand)

9 Performance Summary

Metric	Best Zone	Median Zone	Worst Zone	Average(25 zones)
R^2 (Test)	0.9676	0.9287	0.8031	0.9223
MAE (Test)	18.37	11.31	31.94	15.70
RMSE (Test)	28.17	16.83	47.01	23.54
Relative MAE	0.0918	0.1154	0.1936	0.1257

Table 1: Out-of-sample performance across selected zones

10 Time Series Forecasting Results

10.1 Best Performing Zone

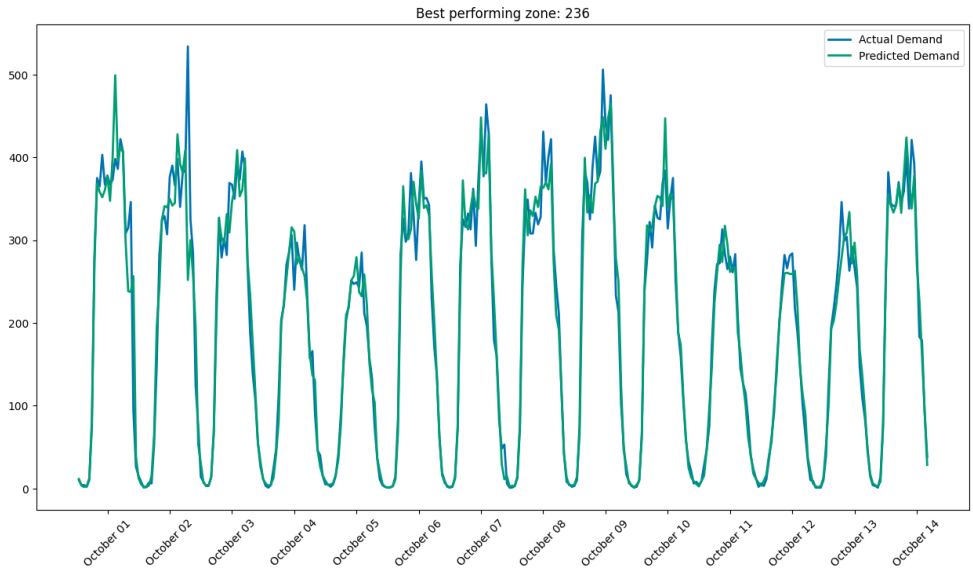

Figure 3: Actual vs Predicted demand (Best Zone)

10.2 Median Performing Zone

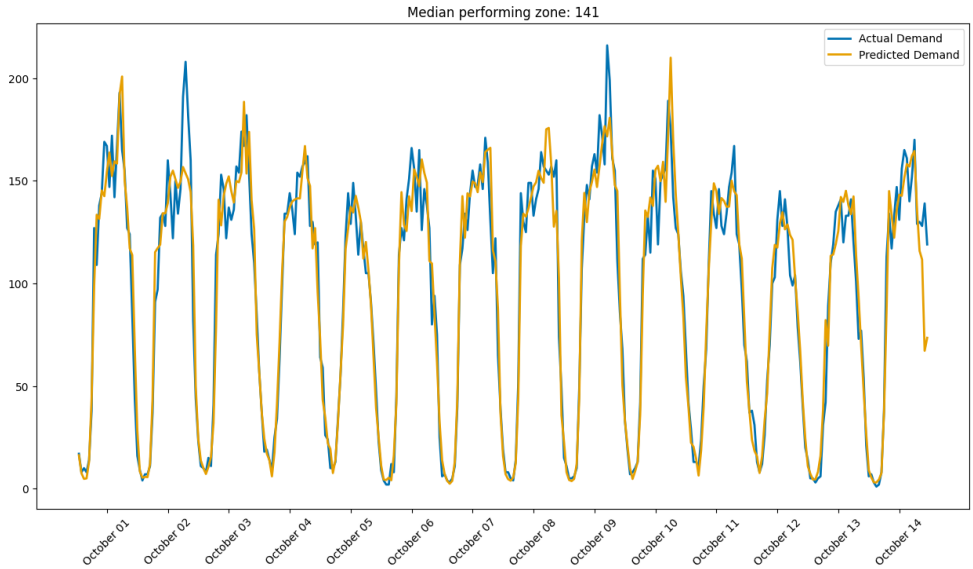

Figure 4: Actual vs Predicted demand (Median Zone)

10.3 Worst Performing Zone

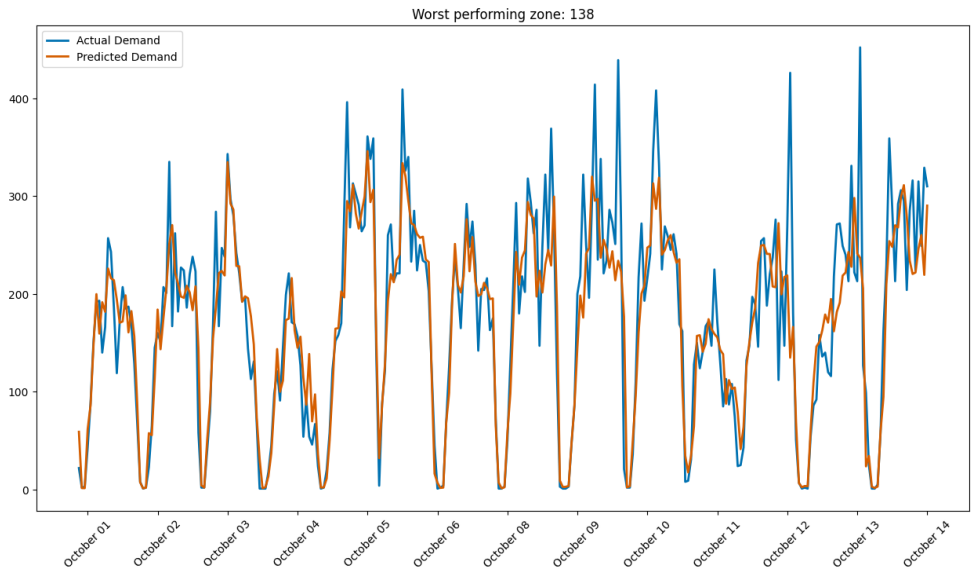

Figure 5: Actual vs Predicted demand (Worst Zone)

11 Residual Analysis

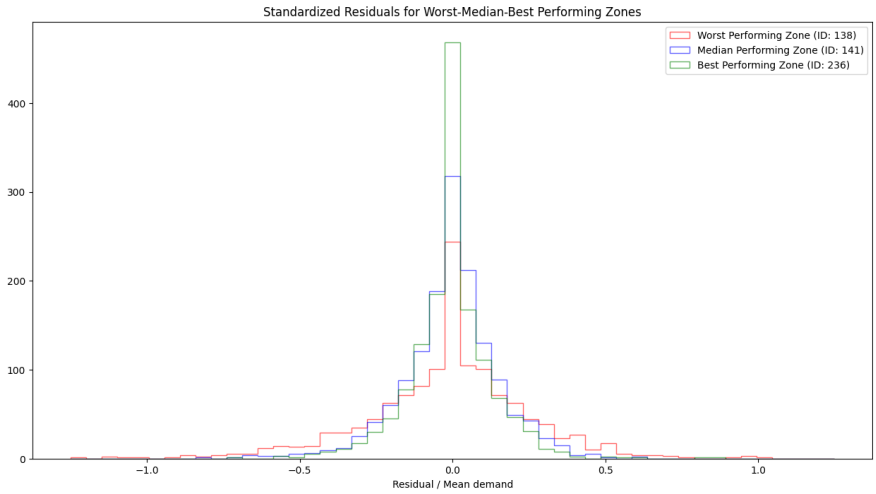

Figure 6: Residual distribution for representative zone

Key observations:

- Residual variance increases during peak demand hours.
- High-volume zones show tighter error distributions.
- Low-volume zones exhibit greater relative volatility.

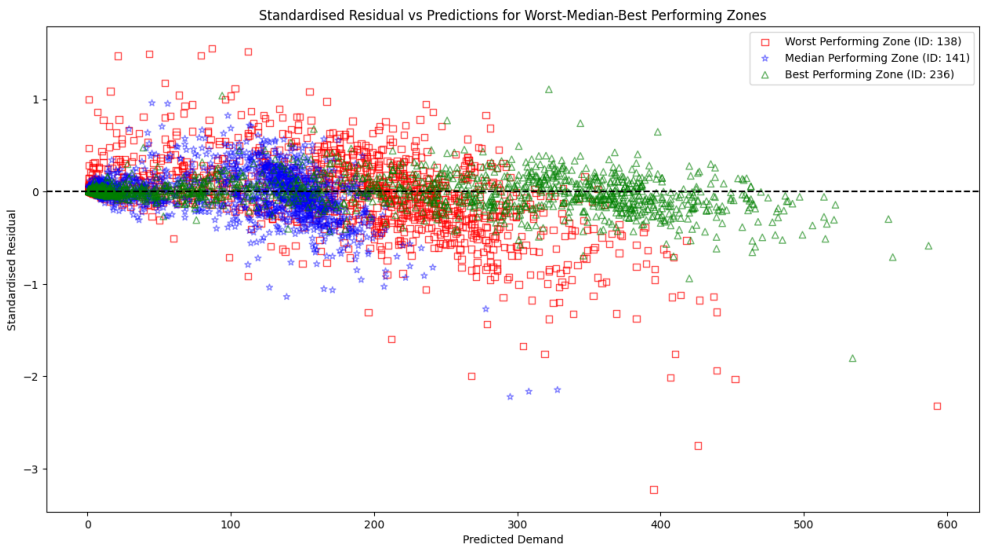

Figure 7: Standardized Residuals vs Predicted Demand for representative zone

Key observations:

- Best-performing zone (green): Residuals are tightly concentrated around zero and uniformly distributed across predicted demand levels, indicating stable and unbiased predictions.
- Median-performing zone (blue): Residuals exhibit moderate dispersion and are primarily concentrated around predicted demand values of 150 trips, suggesting reasonable accuracy in mid-demand regimes.
- Worst-performing zone (red): Residuals show substantial dispersion overall. Importantly, for predicted demand values above 300 trips, all standardized residuals are negative. This indicates consistent overprediction in high-demand scenarios for this zone, pointing to systematic model bias rather than purely random error.

The contrast in residual behavior across zones indicates that a single global model may not fully capture zone-level heterogeneity, suggesting that hierarchical modeling or zone-specific segmentation could improve robustness.

12 Feature Importance Insights

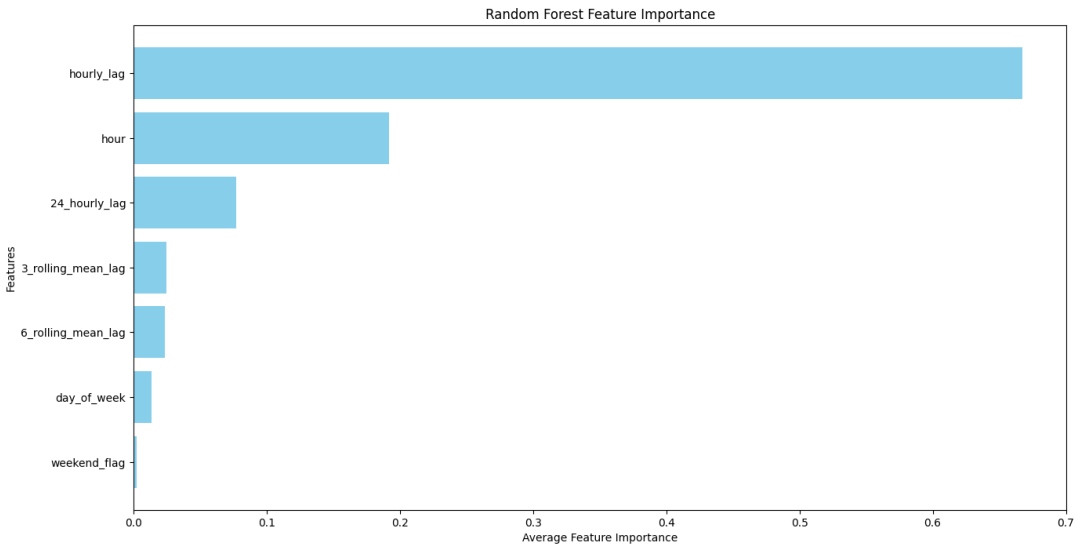

Figure 8: Feature Importances

Feature importance analysis reveals that:

- Hourly lag consistently dominates in high-volume zones.
- Hour-of-day feature encodes structured demand cycles.
- 24 hourly lag captures strong daily seasonality.

13 Computational Considerations

- Raw dataset after cleaning pre-aggregation: 31.86 M rows.
- Filtered dataset to consider pickup zones having more than 500k annual trips.
- Aggregated dataset for selected 25 zones: 800k rows; covers 23.5M trips (74% of total trips from all zones).
- Model artifacts stored via serialization for reproducibility.

14 Limitations

- No exogenous variables (weather, events).
- Independent zone modeling ignores spatial correlations.
- Performance degrades in sparse-demand zones.

15 Future Extensions

- Incorporation of weather and event data.
- Spatial graph-based modeling.
- Gradient boosting or probabilistic forecasting.
- Real-time recursive prediction pipeline.

16 Conclusion

This project demonstrates that structured temporal feature engineering combined with ensemble regression can achieve strong generalization performance in large-scale urban demand forecasting. The framework is computationally efficient, scalable across zones, and adaptable for real-time deployment scenarios.